LOCALIZATION · ACCESSIBILITY

LocaLens — Localization QA Tool

A tool that measures how much translated UI text grows and flags exactly which strings will break their layout, before a translation ships.

ROLE	TIMELINE	INDUSTRY
Designer & Builder	Jun – Jul 2026	Localization / QA Tools

01

Context

Most translation tools stop at the text — they translate a string but say nothing about whether the translated version still fits the button, label or card it was designed for. Teams typically only discover layout breakage after a translated build ships. LocaLens set out to catch that earlier, from a screenshot alone.

02

The Challenge

The product needed to go from a UI screenshot to a reliable overflow risk assessment — extracting text, translating it, and measuring whether the result still fits its original container — without requiring access to the actual codebase or design file.

03

Research

Researched how much text length typically expands or contracts across common target languages — German and Finnish tend to expand, Japanese and Chinese often contract — and used that as the baseline for what counts as a meaningful overflow risk versus normal variation.

- Domain research into text expansion by language
- Competitive audit of translation and localization tools
- Iterative testing of OCR extraction accuracy

04

Insights

The key findings that shaped the direction.

01

Translation length, not translation quality, breaks layouts

A perfectly accurate translation can still break a UI if it's 40% longer than the source string — length risk and translation quality are separate problems that most tools conflate.

02

OCR position data is what makes the check useful

Extracting the text alone isn't enough — knowing where each text block sits and how wide its container is is what turns a translation into an actual overflow measurement.

03

Privacy was a precondition, not a feature

Teams testing pre-release UI screenshots won't upload them to a third-party server — client-side OCR was the only way to make the tool usable for real, unreleased work.

05

Design Strategy

How decisions were made.

01

Start from a screenshot, not a codebase

Built the tool around OCR extraction from a UI screenshot, so it works on any interface — shipped or in-progress — without needing design file or code access.

02

Measure expansion, not just translate

Calculated a character-expansion percentage per text block and compared rendered width against the original container, turning 'is this translated' into 'does this still fit.'

03

Flag risk in three tiers

Colour-coded every text block as Safe, Review or High-risk, so a team can triage a screen with dozens of strings in seconds instead of reading every translation.

04

Keep it client-side

Ran OCR entirely in the browser so screenshots of unreleased UI never leave the user's machine, making the tool safe to use on pre-launch work.

06

Solution

A client-side localization QA tool that extracts text from a UI screenshot via OCR, translates it into 10 languages, and flags exactly which strings risk breaking their original layout.

- ✓ OCR extraction of text blocks with position data from PNG, JPG or WEBP screenshots
- ✓ Translation into 10 languages including French, German, Japanese and Arabic
- ✓ Character-expansion and container-overflow risk scoring per text block

- ✓ Side-by-side visual comparison with zoom and pan, exportable as PNG, PDF, CSV, XLSX or JSON

07

Design System Thinking

This project wasn't designed as a one-off — here's how it connects back to the system.

Components

A reusable risk-badge component (Safe / Review / High-risk) applied consistently across the comparison view and exported reports.

Tokens

A three-tier risk colour scale shared between the live comparison view and every export format.

Patterns

A side-by-side comparison pattern with synchronized zoom and pan across source and translated versions.

Governance

Fully client-side processing — screenshots and extracted text never leave the browser, by design rather than by policy.

08

Impact

LocaLens shipped as a working QA step most localization workflows skip entirely, turning 'ship the translation and see what breaks' into a check a team can run before release.

10	3	100%
Languages supported	Risk tiers: Safe, Review, High-risk	Client-side — no server upload

09

Reflection

This was the clearest case of a tool needing a genuinely different data model than I expected going in — I started thinking of it as a translation tool and only later realised the actual product was a measurement tool that happens to use translation as an input.